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Wireless technologies use electromagnetic waves to transmit data between devices without the need for physical connections.

Different types of wireless technologies:

1. Bluetooth

A short-range technology that allows devices to connect and exchange data, such as for audio transmission and smart home automation.

Applications:

Wireless Audio: Headphones, speakers, and wireless earbuds.

Personal Area Networks (PAN): Connecting devices like smartphones, tablets, and laptops.

IoT Devices: Smartwatches, fitness trackers, and health monitors.

2. WiFi

A common networking technology that allows users to connect to a router within close proximity.

Applications:

Internet Access: Commonly used in homes, businesses, and public places for high-speed internet.

Streaming: HD video and music streaming (e.g., Netflix, YouTube).

Office Networking: For connecting computers, printers, and servers.

Gaming: Online gaming and multiplayer connections.

3. Infrared communication

Uses IR radiation to transmit information over short distances, such as for TV remote controls and security control.

Applications:

Remote Controls: TVs, air conditioners, sound systems, and other home appliances.

Point-to-Point Communication: Data transfer between devices like printers and computers.

Healthcare: Infrared thermometers for non-contact temperature measurement.

Security Systems: Motion detectors and infrared sensors for surveillance systems.

4. Cellular

A reliable broadband technology that supports voice calls and streaming applications.

Applications:

Mobile Communication: Voice and data services on smartphones and mobile devices.

Mobile Internet: High-speed internet access on mobile phones, tablets, and laptops.

IoT: Connecting IoT devices in smart cities, agriculture, and healthcare.

Telemedicine: Remote health monitoring and consultations.

5.Satellite communication

Uses satellites to allow people around the world to communicate with each other.

Applications:

Global Communication: Providing internet and phone services in remote and underserved areas.

Navigation: GPS and satellite-based navigation for vehicles, ships, planes, and mobile apps.

Broadcasting: TV and radio broadcast transmission globally.

Weather Forecasting: Monitoring weather patterns and natural disasters from space.

6.LoRa

A low-power wireless technology that can transmit data over long distances, such as for IoT devices.

Applications:

Smart Cities: Traffic monitoring, waste management, and environmental sensing.

Agriculture: Remote monitoring of crops, soil, and weather conditions.

Asset Tracking: Monitoring the location of goods in transit over large distances.

Environmental Monitoring: Tracking air quality, water levels, and pollution.

7.RFID

Uses radio waves to identify objects or people.

Applications:

Supply Chain Management: Tracks inventory and goods in real-time across the supply chain.

Retail & Asset Management: Prevents theft, manages stock, and authenticates products.

Access Control & Security: Provides secure building access, event ticketing, and parking management.

Healthcare & Logistics: Tracks patient IDs, medical equipment, luggage, and vehicles.

Comparison between different wireless technologies based on different parameters

Technology	Range of Communication	Cost of Devices	Propagation Delay	Power Consumption	Throughput	Fan Out
Bluetooth	~10 meters (Classic), ~100 meters (BLE)	Low (Affordable consumer devices)	Very low (~1 ms)	Low	~1-3 Mbps (Classic), ~2 Mbps (BLE)	Limited (7 active connections, typically up to 50 passive devices for BLE)
Wifi	~50 meters (indoors), ~200	Moderate	Low (~1-10 ms)	Medium-High	Up to ~9.6 Gbps (Wi-Fi)	High (Dozens of

	meters (outdoors)				6E)	devices per access point, dependent on router capacity)
Zigbee	~10-100 meters (line-of-sight)	Low	Low (~10 ms)	Very low	~250 kbps	Moderate (~65,000 devices in a mesh network, practical deployment usually less)
LoRa	~10-15 kilometers (rural), ~2-5 km (urban)	Moderate	Moderate (~10-50 ms)	Very low	~0.3-50 kbps	High (Up to thousands of devices in a network)
5G	~100 meters to several kilometers (depends on spectrum)	High	Very low (<1 ms)	High	Up to ~10 Gbps	Very High (Massive device connectivity, supporting IoT and dense urban scenarios)
NFC	~10 cm	Low	Very low (<1 ms)	Very low	~424 kbps	Limited (1-to-1 connection, no fan-out)
Infrared	~1-5 meters (line-of-sight)	Low	Very low (<1 ms)	Low	~1-4 Mbps	Very limited (point-to-point communication only)
Satellite Communication	Global coverage	High (expensive infrastructure)	High (~250 ms for GEO)	High	~100 Mbps (modern satellites)	Moderate (Varies depending on system, can support hundreds to thousands of concurrent users)

Cellular	~1-10 km (varies with base stations and frequency)	High	Low (~10-50 ms for 4G, <10 ms for 5G)	Medium-High	~1 Gbps (4G), ~10 Gbps (5G)	High (Supports thousands of users per cell tower)